



Index Number:

NATIONAL CERTIFICATE OF EDUCATION

2025

TECHNOLOGY STUDIES (N550)

TIME: 1 HOUR 15 MINUTES

Component 1

Candidates answer on the Question Paper.

Additional materials: Standard drawing equipment

READ THESE INSTRUCTIONS FIRST

- Write your index number in the space provided above and on the A3 printed question sheet.
- Write in dark blue or black ink. Coloured and HB pencils may be used for diagrams/sketches and shading.
- Answer **ALL** questions.
- Check whether this assessment booklet consists of:
 - Questions **1 - 7** printed on **8** pages from page **2** to page **9**;
 - Question **8** printed on an **A3** sheet of paper as an insert.
- Do not use correction fluid, highlighters or glue.
- All answers must be written in the spaces provided.
- The total marks for each question is given at the beginning of each question.
- The number of marks for each item is shown in brackets [].
- Question 8:**
 - All drawings must be accurately drawn using drawing instruments.
 - All construction lines **MUST** be shown.
 - All dimensions are in millimetres.
- The total number of marks for this paper is **50**.
- At the end of the examination, fold the A3 sheet and insert it in the assessment booklet.**

For Examiners' use											
Question No.	1	2	3	4	5	6	7	8(a)	8(b)	Total	Signature
Examiner											
Team Leader											
CE/ACE											

SECTION A**Question 1 (5 marks)**

Circle the correct answer. Each item carries one mark.

1. Which tool is shown in **Figure 1**?

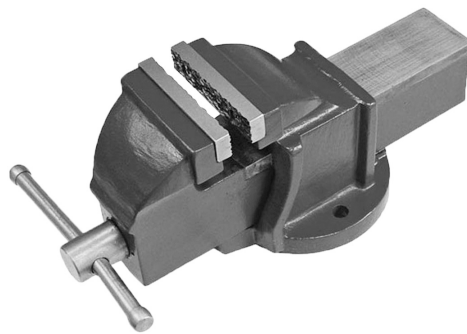


Figure 1

- | | | | |
|----------|--------------|----------|-----------------|
| A | Bench vice | B | Bench hook |
| C | Machine vice | D | Engineer's vice |
2. Which of the following is an example of a thermosetting plastic?
- | | | | |
|----------|-----------------|----------|--------------------|
| A | Polyester resin | B | Polypropylene |
| C | Polyamide | D | Polyvinyl chloride |

3. Which mechanism is shown in **Figure 2**?

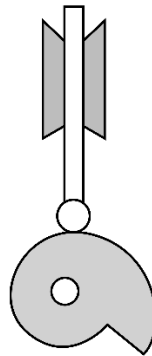


Figure 2

- | | | | |
|----------|--------|----------|---------|
| A | Pulley | B | Cam |
| C | Lever | D | Linkage |

4. Which manufactured board is made of wood fibres mixed with resin and wax?

A	Blockboard	B	Chipboard
C	Plywood	D	Medium Density Fibreboard (MDF)

5. Which of the following tool is used to mark parallel lines to the face side or the face edge of a wooden workpiece?



Question 2 (5 marks)

Put a tick (✓) in the appropriate column to indicate whether the statements in the table below are **True** or **False**. An example is given.

Statement		True	False
Example: <i>A hacksaw is used to cut metals.</i>		✓	
(a)	Hydraulic systems use pressurised air to transmit motion and forces.		
(b)	Rotary motion follows a circle.		
(c)	Solar panels convert wind energy into electrical energy.		
(d)	Acrylic is an eco-material.		
(e)	Biodegradable materials are environmentally friendly.		

E

TL

CE

Question 3 (5 marks)

Fill in the blanks with the appropriate word(s) from the list given below.
An example is given.

Ferrous	Rigid polyurethane	Non-ferrous	Brass
design brief	Mahogany	Epoxy resin	

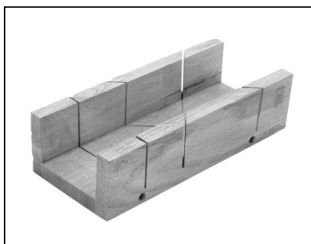
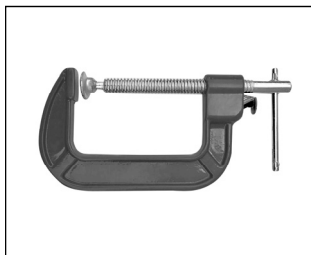
Example: A **design brief** is a short statement of intent indicating what is to be designed.

- (a) is used to make floats.
- (b) is a hardwood.
- (c) metals are attracted to magnets.
- (d) is resistant to corrosion from salt water.
- (e) has good adhesive qualities.

Question 4 (5 marks)

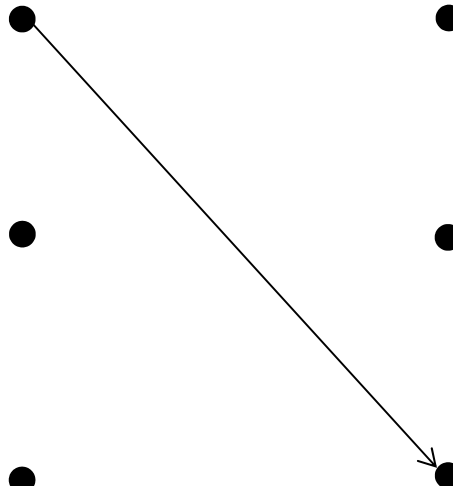
Match each tool in **Column A** to its name in **Column B**.
An example is given.

Column A



Column B

- Scriber
- Sliding bevel
- Hand drill
- Mitre box
- Try-square
- Metal file
- G-clamp

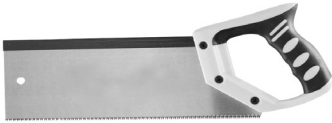


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Question 5 (6 marks)

- (a) Name each cutting tool in the table below. Choose from the given list.
An example is given.

Hacksaw	Hand saw	Coping saw	Tenon saw	Junior hacksaw
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Cutting tool	Name
Example: 	Tenon saw
	
	
	

[3]

- (b) Sketch an **engineer's square** in the space provided below.

[3]

SECTION B**Question 6 (4 marks)**

Figure 3 shows an isometric drawing of a shaped block.

Apply the thick and thin line technique to the shaped block to enhance it.

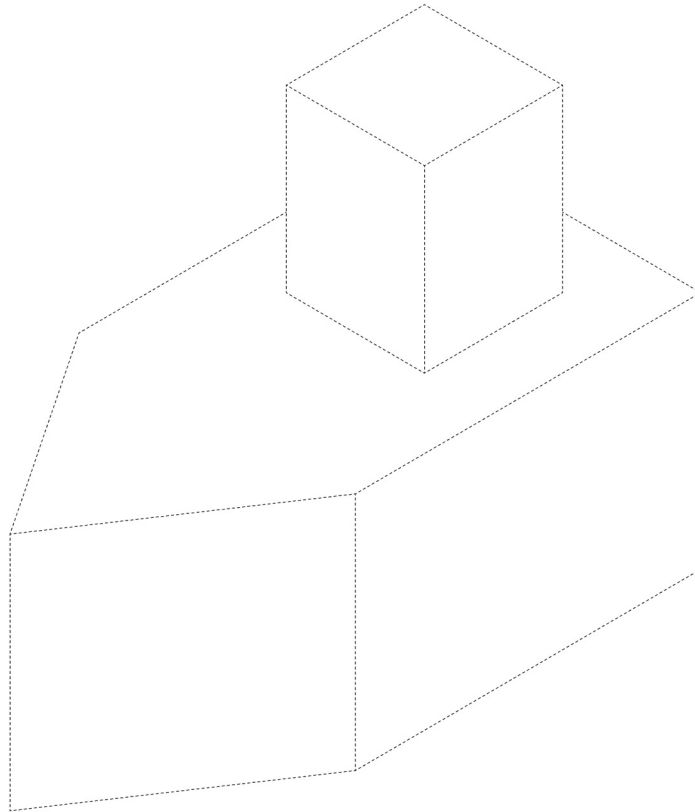


Figure 3: Shaped block

Question 7 (5 marks)

A glass water bottle is shown in **Figure 4**. A carrier is required to carry **two** glass water bottles securely.

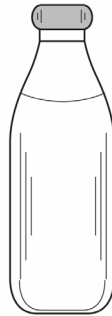


Figure 4: Glass water bottle

In the space provided, sketch **one** idea for the carrier to hold the two glass water bottles and complete the analysis for the material(s) used and its safety.

Material(s) used:	Safety:
.....	
.....	

For Question 8
Please see A3 insert

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